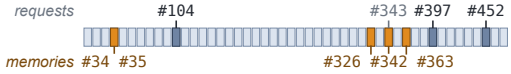


Group conversations

50 groups · 80,299 messages

de-identified, real-world everyday settings
Chinese + English translation

Trip-planning group
2,236 turns
10 messages per cell



1 Locate requests

13,782 candidates → 7,926 kept

a Enumerate candidate requests
b Answer with and without history

#397 @Agent check when we can start booking tickets

with You have an Apr 30 train and start watching…
without Please confirm the booking time…
history changes the answer ✓keep

#343 @Agent check the rail fare from Chongqing to Chengdu…

with About ¥100–200 for second class; check 12306.
without About ¥100–200 for second class, …
same answer without it ✗drop

2 Anchor memories

22,843 anchors, 154 rejected

a Extract memories
for request #397

anchor *a = (f, t, e)*
fact departure moved to Apr 30
turn **#342** Zhou Yue
span "we leave on Apr 30"

#34 meet 7:30 am, May 1
#326 rail both ways

b Verify evidence spans
#342 verbatim ✓keep
#431 not found ✗drop

Verified anchors go to a per-group inventory.

3 Assign type

595 pairs → 338 kept

a Count units
M1 one memory unit
M2 two or more
M3 M2 + conflict pair
#34 ~~May 1~~ *old*
#342 Apr 30 *new*

b Detect updates
c Three votes
same matter? ✓✓✓
d Current at cut
if not, the cut moves before the next change

4 Build instances

instance (c, q, A, K)

a Write authored queries
b Derive facts, grounding
c Trigger penalties
d Propose service criteria

A L1 × M2 *native*
#104 @Agent on departure day, remind…
meet 7:30 am, May 1 #34
no-show pays ¥500 #35

B L2 × M3 *authored*
#452 @Agent Post a reminder…
+3 window seat #363
+2 leaves Apr 30 **#342**
−2 uses old May 1 #34

−2 **superseded value**

5 Quality control

9,542 pass, 440 fail

a Rule gates
✓ anchors precede the cut
✓ spans found verbatim
✓ query gives no cue of a change

b Model review
✓ each anchor is necessary
✓ query not answerable from its own wording
✓ assistant turns only relay a member's fact
✓ M3 values concern one matter

c Human validation
3 annotators, 120 instances: 95.8% valid by majority

Evaluation set

2,395 instances
sampled from the 9,542

Penalty criteria from real feedback

#1474 "I asked you to keep the memo. How is it mixed up again?"

423 negative reactions, **97 corrections**

five failure patterns, each with its weight

−2 recalled fact · −1 settled plan · −2 agreement or constraint · −3 member's current state

rules · language model · humans · kept / dropped · anchor = required memory · judge only · *(c, q, A, K)* cutoff, query, required memories, answer key